



Task :- 2

Submitted to

Code Alpha

Submitted by

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Submit :- code + Circuit Diagram +Output Image

1.Code

```
digitalWrite(Pump_Pin,HIGH);
lcd.setCursor(10,3);
lcd.print(" Motor:ON ");
}
else
{
digitalWrite(Pump_Pin,LOW);
lcd.setCursor(10,3);
lcd.print(" Motor:OFF ");
}
}

{
lcd.setCursor(10,1);
lcd.print(" R=YES ");
}

//Soil Moisture
int S3=analogRead(A2); // Read Soil Moisture
int SM=S3/10;
lcd.setCursor(0,2);
lcd.print(" S=");
lcd.print(SM);
lcd.print("% ");

//Air Humidity
int S4=analogRead(A3); // Read Air Humidity
int H=S4/10;
lcd.setCursor(10,2);
lcd.print(" H=");
lcd.print(H);
lcd.print("% ");

if(Temp>30)
{
digitalWrite(Fan_Pin,HIGH);
lcd.setCursor(0,3);
lcd.print(" Fan:ON ");
}
else
{
digitalWrite(Fan_Pin,LOW);
lcd.setCursor(0,3);
lcd.print(" Fan:OFF ");
}

if(SM<40 && R==0)
{
// include the library code:
#include <LiquidCrystal.h>

//LiquidCrystal lcd(rs, en, d4, d5, d6, d7);
LiquidCrystal lcd(13, 12, 11, 10, 9, 8);

const int Fan_Pin = 6;
const int Pump_Pin = 7;

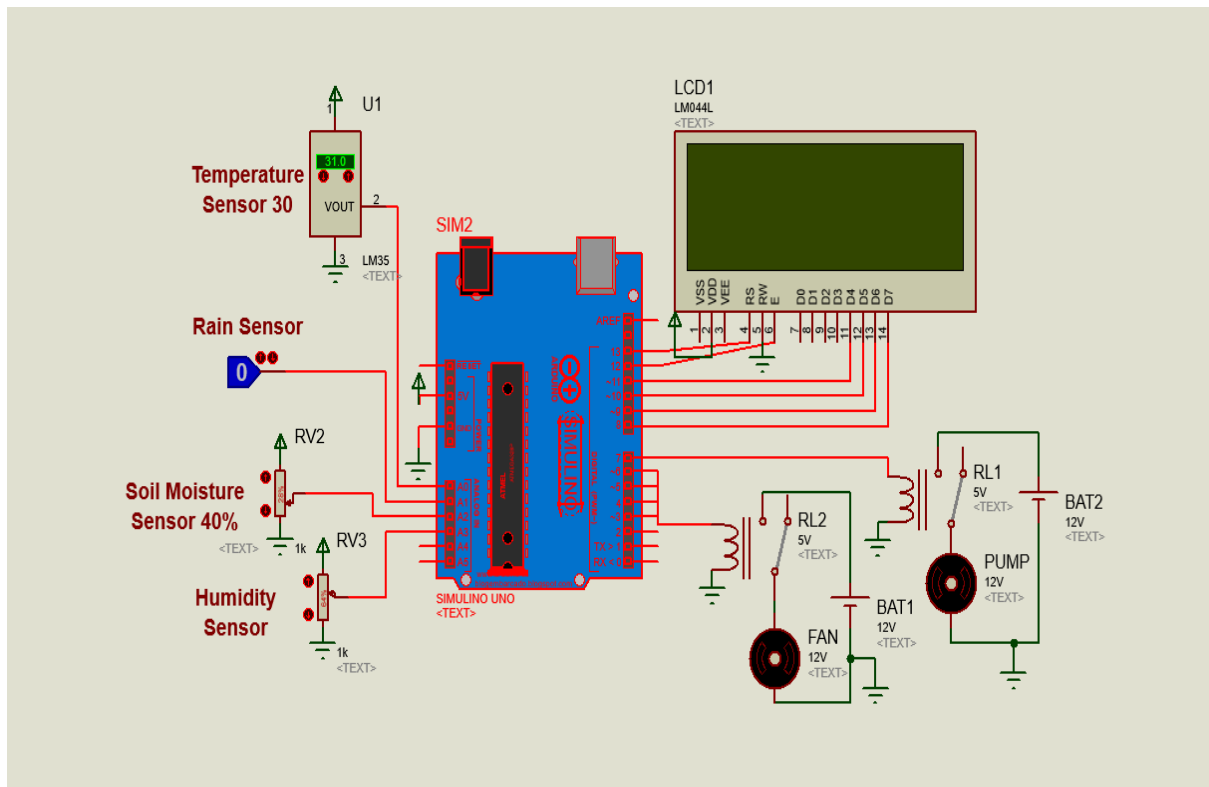
void setup()
{
lcd.begin(20, 4); // set up the LCD's number of columns and rows:
lcd.setCursor(0,0); // set the cursor position:
lcd.print(" IoT SMART FARMING ");

pinMode(Fan_Pin,OUTPUT);
pinMode(Pump_Pin,OUTPUT);
pinMode(A1,INPUT);
}

void loop()
{
//Temperature Sensing
int S1=analogRead(A0); // Read Temperature
float mV=(S1/1023.0)*5000; // Storing value in Degree Celsius
int Temp = mV/10;
lcd.setCursor(0,1);
lcd.print(" T=");
lcd.print(Temp);
lcd.print("°C ");

//Rain Sensing
int R=digitalRead(A1); // Read Rain Sensor
if(R==0)
{
lcd.setCursor(10,1);
lcd.print(" R=NO ");
}
```

2.Circuit Diagram



3.Output Image

